

MBIM — Memory Boundary Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-MEM-MBIM-2026-06-08-0003

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Memory Integrity Governance Layer

Governed Space: Memory Boundary Integrity

Category: AI & Interpretation

Subcategory: Memory Boundary Governance

Type: Memory Integrity Module

Parent Standard: Memory Integrity Standard (MIS)

Version: 1.0

Status: Canonical · Open Module

Effective Date: 8 June 2026

Compatibility: OOF Methodology OS · Memory Integrity Standard (MIS) ·

Memory Continuity Integrity Module (MCIM) · Memory Traceability Integrity Module (MTIM) ·

User Memory Integrity Standard (UMIS) · Memory Governance Standard (MGS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Memory Boundary Integrity Module (MBIM) defines the structural conditions under which memory boundaries remain identifiable, governable, protected, accountable, and operationally valid throughout memory creation, storage, access, modification, sharing, and utilization processes.

MBIM governs memory boundaries.

The module ensures that memory remains properly separated, classified, protected, and governed according to its ownership, purpose, authority, context, sensitivity, and operational role.

A system satisfies MBIM only if:

- memory boundaries remain identifiable
- memory ownership remains visible
- access boundaries remain governable
- memory separation remains preservable
- boundary violations remain detectable
- memory boundaries remain operationally valid

A memory environment that cannot determine where one memory domain ends and another begins does not satisfy MBIM.

Module Operational Role

MBIM defines the memory-boundary governance layer of MIS.

Its role is to preserve valid separation and governance of memory domains.

Module Operational Space

MBIM governs:

- memory boundaries
- memory ownership
- memory separation
- memory classification
- memory access domains
- memory authority domains
- boundary governance
- boundary accountability

The module applies wherever memory must remain separated according to governance requirements.

Module Function

The module applies wherever systems must preserve:

- memory ownership integrity
- memory-domain separation
- governance-valid access boundaries
- accountable memory sharing
- protected memory environments
- operationally reliable memory governance

Its function is to ensure that memory remains contained within valid governance boundaries.

Minimum Implementation Framework

1. Define the Memory Boundary Object

The organization must define which memory domains require boundary governance. This may include:

- user memory
- agent memory
- organizational memory
- institutional memory
- Human-AI memory
- private memory
- shared memory
- operational memory

2. Define Memory Boundary Conditions

The system must define the conditions under which memory boundaries remain valid.

This includes:

- ownership requirements
- access requirements
- authority requirements
- separation requirements
- classification requirements
- governance-valid boundary conditions

3. Define Boundary Degradation Detection Logic

The system must define how boundary degradation is identified.

This may include:

- unauthorized access
- memory leakage
- ownership ambiguity
- cross-domain contamination
- authority violations
- invalid memory sharing

4. Define Operational Response or Governance Logic

The system must define governance logic for boundary-integrity failures.

Governance response may include:

- boundary review
- access restriction
- governance intervention
- escalation
- memory isolation
- containment procedures
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- boundary definitions
- ownership assignments
- access activities
- governance reviews
- intervention actions
- resulting memory states

A memory environment must not remain boundary-valid if materially significant memory boundaries cannot be identified, enforced, reviewed, or governed.

Use Case 1 — Multi-Agent AI Environment

Scenario

Multiple AI agents operate within a shared ecosystem while maintaining separate memory domains, permissions, responsibilities, and operational roles.

Application

MBIM governs memory ownership, separation, access boundaries, and cross-agent memory governance.

Result

The environment gains stronger security, reduced memory contamination risk, and improved governance clarity.

Use Case 2 — Human-AI Enterprise Platform

Scenario

An enterprise platform manages employee memory, organizational knowledge, customer information, and AI memory environments simultaneously.

Application

MBIM governs memory separation, ownership integrity, authority boundaries, and controlled information sharing.

Result

The organization gains stronger governance reliability, reduced boundary violations, and improved operational trust.

Canonical Closing Statement

Memory Boundary Integrity Module (MBIM) defines the structural conditions under which memory boundaries remain identifiable, governable, protected, accountable, and operationally valid throughout memory creation, storage, access, modification, sharing, and utilization processes.

Memory cannot remain governable if its boundaries disappear. Memory boundary integrity therefore becomes a foundational integrity condition of governable memory systems.

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Canonical Source

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